

KPRO PIPELINE DIRECTORY STRUCTURE

Simplified structure for the Kaleidoscope Pro data pipeline

Updated: 2025-12-29

ACTUAL PROJECT STRUCTURE

```
Kpro_Masterfile_Pipeline/
|
|— R/                # All R function modules
| |— functions/
| | |— load_all.R    # Master loader
| | |— core/         # utilities.R, schema_detection.R, config.R
| | |— ingestion/    # ingestion.R
| | |— standardization/ # standardization.R, datetime_conversion.R
| | |— analysis/     # callspennight.R, detector_mapping.R, etc.
| | |— validation/   # validation.R
| | |— output/       # visualization.R, summarization.R
| |— workflows/      # Main workflow scripts
| | |— 01_ingest_raw_data.R
| | |— 02_standardize.R
| | |— 03_generate_cpn_template.R
| | |— 04_finalize_cpn.R
|
|— data/             # Input data (ALL GITIGNORED)
| |— raw/            # User's local CSV files (personal data)
| |— sample_data/    # Example data for public release
| | |— example_id.csv # Small sample for testing
|
|— outputs/          # INTERMEDIATE files (GITIGNORED)
| |— 01_intro_standardized/ # After ingestion, before standardization
| | |— raw_combined_YYYYMMDD_HHMMSS.csv
| |
| |— 03_templates/   # CallsPerNight templates
| | |— CallsPerNight_Template_ORIGINAL_*.csv
| | |— CallsPerNight_Template_EDIT_THIS_*.csv
| |
| |— 04_edit_logs/   # Edit tracking
| | |— CallsPerNight_EditLog_*.txt
|
|— results/          # FINAL outputs (GITIGNORED until public release)
| |— csv/            # ✅ ALL FINAL DATASETS GO HERE
| | |— kpro_master_YYYYMMDD_HHMMSS.csv # Final master from Workflow 02
| | |— CallsPerNight_final_v*.csv     # Final CPN from Workflow 04
| |
| |— plots/          # All visualizations
```

```

| | | └─ activity/      # Activity patterns
| | | └─ diagnostics/  # Data quality plots
| | | └─ effort/       # Recording effort heatmaps
| | |
| | └─ tables/         # Summary tables (GT, kable, CSV)
| |
| └─ report/           # Rendered reports (Quarto/RMarkdown)
|   └─ analysis_report.html
|
└─ docs/               # Documentation (NOT gitignored)
  | └─ diagrams/       # Mermaid flowcharts
  |   └─ workflow_overview.mmd
  |   └─ schema_detection.mmd
  |
  └─ CODING_STANDARDS.md # Development guidelines
|
└─ inst/               # Installation files (NOT gitignored)
  | └─ config/
  |   └─ study_parameters.yaml
  |   └─ YAML_PARAMETER_GUIDE.md
  |
  └─ logs/              # Execution logs (GITIGNORED)
    | └─ pipeline_log_*.txt # Workflow execution logs
    | └─ error_log.txt     # Error tracking
    | └─ audit_report.txt  # Standards compliance
    |
    └─ tests/            # Unit tests (NOT gitignored)
      | └─ test_master.R
      | └─ test_functions.R
      | └─ test_*.R
      |
      └─ scripts/        # Standalone utilities (NOT gitignored)
        | └─ audit_code.R # Standards checker
        |
        └─ .gitignore     # Git exclusions (see below)
        └─ README.md      # Project overview
        └─ Kpro_Masterfile_Pipeline.Rproj

```

GITIGNORE STRATEGY

Current Phase (Development with Personal Data)

gitignore

```
# Data directories (personal bat data)
data/raw/
data/external/

# Intermediate outputs (checkpoints, templates with personal data)
outputs/

# Final results (all contain personal data)
results/

# Logs (may contain file paths with personal info)
logs/

# R session data
.RData
.Rhistory
.RProfile

# RStudio files
.Rproj.user/
```

Future Phase (Public Release)

After removing personal data and adding sample data:

```
gitignore

# Keep sample data visible
# data/sample_data/ # NOT ignored

# Show example outputs (with sample data only)
# outputs/ # Can selectively un-ignore
# results/ # Can selectively un-ignore

# Always ignore user data directories
data/raw/
data/external/

# Always ignore logs
logs/

# Always ignore R session files
.RData
.Rhistory
```

DIRECTORY PURPOSES

Directory	Purpose	For Whom	Timeline
R/	Function library	Developers	Public now
data/raw/	User's personal CSVs	Individual users	Never public
data/sample_data/	Example data for testing	Everyone	Public later

Directory	Purpose	For Whom	Timeline
outputs/	Pipeline checkpoints	Developers	Gitignored for now
results/	Final deliverables	End users	Public later (with samples)
docs/	Documentation	Everyone	Public now
inst/config/	Configuration templates	Users	Public now
logs/	Execution tracking	Developers	Never public
tests/	Unit tests	Developers	Public now
scripts/	Utility scripts	Developers	Public now

KEY DESIGN DECISIONS

1. No data/external/ folder

Why: External data paths are defined in YAML, not in project structure. Users specify paths like `F:/bat_data/` in `study_parameters.yaml`.

2. No outputs/02_standardized/

Why: The standardized master file IS the final output. It goes directly to `results/csv/kpro_master_*.csv`

3. CallsPerNight saves to results/csv/

Why: This is a FINAL deliverable, not an intermediate checkpoint. Users want it in results/ alongside the master file.

4. Everything gitignored during development

Why: All current data is personal bat research data. Will selectively expose directories with sample data before public release.

DEFERRED DECISIONS

1. Renaming outputs/01_intro_standardized/

- Current name works, not a priority
- Could rename to `01_raw_combined/` or `01_ingested/` later

2. Template organization in outputs/

- Currently scattered in outputs/ root
- Could create `outputs/03_templates/` and `outputs/04_edit_logs/` later
- Not urgent since outputs/ is gitignored anyway

SAMPLE DATA STRUCTURE (For Future Public Release)

```
data/sample_data/
├── README.md      # Explains sample data origin
├── detector_001_id.csv # Small example (100-200 rows)
```

```
|— detector_002_id.csv # Different schema version
|— detector_003_id.csv # Mixed schema example

# Paired with sample results:
results/csv/
|— sample_master.csv # Expected output
|— sample_CallsPerNight_v1.csv # Expected output
```

Sample data requirements:

- De-identified (no real detector IDs or locations)
- Small (few hundred rows total)
- Demonstrates all 3 schema versions (v1, v2, v3)
- Shows realistic edge cases (NoID calls, overnight recording, etc.)
- Includes detector mapping template example

FINAL NOTES

Current state: All working directories gitignored (personal data protection)

Before public release:

1. Remove all personal data
2. Add sample data to data/sample_data/
3. Run pipeline with sample data to generate example outputs
4. Selectively un-ignore results/ to show expected outputs
5. Update README with sample data workflow

Philosophy:

- `outputs/` = Working directory for developers
- `results/` = Final deliverables for end users
- Both gitignored now, selectively exposed later with samples